Seeed Studio XIAO ESP32C6 Rescue Node Pinout

Custom Rescue Node Architecture: Specifically configured for the **Seeed Studio XIAO ESP32C6** (ESP32-C6FH4) with IndieSemiC SX1262 LoRa, u-blox NEO-6M GPS, MPU6050, external waterproof button, and buzzer/LED.

Pin Assignment Summary (11 Header Pins)

XIAO Pin	ESP32-C6 GPIO	Target Component	Function / Notes
D0	`GPIO 0`	**Piezo Buzzer / Status LED**	3V Buzzer & LED wired in parallel to D0 & GND
D1	`GPIO 1`	**SX1262 LoRa**	BUSY indicator line
D2	`GPIO 2`	**SX1262 LoRa**	DIO1 interrupt line
D3	`GPIO 21`	**SX1262 LoRa**	NSS / SPI Chip Select (CS)
D4	`GPIO 22`	**MPU6050 Accelerometer**	I2C SDA (Data line)
D5	`GPIO 23`	**MPU6050 Accelerometer**	I2C SCL (Clock line)
D6	`GPIO 16`	**Tactile Waterproof Button**	Connect button between D6 & GND
D7	`GPIO 17`	**NEO-6M GPS Module**	RX (Connects to GPS TX only)
D8	`GPIO 19`	**SX1262 LoRa**	SPI SCK (Clock)
D9	`GPIO 20`	**SX1262 LoRa**	SPI MISO (Data in)

XIAO Pin	ESP32-C6 GPIO	Target Component	Function / Notes
D10	`GPIO 18`	**SX1262 LoRa**	SPI MOSI (Data out)
RST / EN	`RST Pad`	**SX1262 LoRa**	LoRa RESET connected to XIAO RST pad (hardware sync)

Hardware Bypasses & Optimization

Optimization	Implementation Detail
SX1262 RESET	Bypassed from GPIO. Connect LoRa RESET directly to the XIAO `RST` pad/pin to reboot together with the MCU.
NEO-6M TX	GPS transmit-only mode. Connect GPS TX to XIAO D7 (GPIO 17). Leave GPS RX floating.
Onboard BOOT Button	The onboard physical button on the XIAO board is tied to `GPIO 9` with a pull-up resistor, serving as an auxiliary user button.

Power & Antenna Connections

Rail / Interface	Connection Details
3.3V Rail	Clean 3.3V from AP2112K LDO to XIAO 3V3, SX1262 VCC, GPS VCC, MPU6050 VCC
GND Rail	Common ground across all modules, external button, and buzzer/LED
LoRa Antenna	865–868 MHz Stub Antenna via IPEX to SMA Female pigtail to SX1262 RF port
GPS Antenna	Active 25x25mm Ceramic Patch Antenna to NEO-6M antenna port
BLE Antenna	Onboard ceramic antenna on the Seeed Studio XIAO ESP32C6

Meshtastic Custom Firmware ESP32-C6 (Seeed Studio XIAO ESP32C6) • Environment: env:rescue_node_c6